
Compact-gauge topology and the classical-statistical vacuum in real-time lattice scalar QED plasmas

Anonymous Author(s)
anonymous@example.com

Abstract

Real-time lattice scalar quantum electrodynamics (QED) is a variational tool for simulating relativistic-quantum plasmas in the strong-field regime, where collective plasma scales overlap intrinsic QED scales. We examine two regimes in which this framework extends cleanly—a two-species electron-ion plasma and geometric-optics wave propagation through an inhomogeneous density profile—and two regimes in which it fails for reasons rooted in its construction rather than in numerical resolution. First, because the gauge field enters through compact Wilson lines, the discrete gauge field lives on a torus: a uniform background vector potential folds the matter-field dispersion with period $2\pi/(q\Delta x)$, an artifact of the compact U(1) topology that no lattice refinement removes. We show that replacing the Wilson line by a non-compact covariant derivative eliminates the folding while preserving global charge conservation to machine precision, at the cost of relaxing exact local gauge invariance. Second, the classical-statistical approximation on which the method rests carries no vacuum fluctuations when initialized from a coherent state: the matter-field response to a supercritical Sauter pulse vanishes linearly with the background density, whereas quantum Schwinger pair production is density-independent. Seeding the Wigner function of the quantum vacuum (half a quantum per mode) restores a finite, seed-independent pair-production response. Both reproductions of the original scheme’s demonstrations are verified, and every conclusion is supported by convergence studies across lattice resolutions. The two failure modes reveal the structural limits of real-time lattice scalar QED and delineate the modifications needed to move beyond them.

1 Introduction

When a plasma is dense or the fields are strong, the kinetic description of a plasma—itsself a geometrical-optics approximation to the underlying relativistic-quantum theory—ceases to be adequate. Electron-positron pair production by intense lasers, wave dynamics on scales comparable to the Compton wavelength, and astrophysical plasmas in extreme magnetic fields all demand a treatment in which quantum and collective scales are no longer cleanly separated [2, 3, 4, 5]. Semiclassical remedies, which insert prefabricated source terms into kinetic or fluid equations [6, 7, 8, 9, 10], fail to capture coexisting QED processes self-consistently and, once many pairs are produced, no longer respect energy-momentum conservation.

A faithful description requires the full relativistic-quantum theory, but path-integral lattice QED is numerically prohibitive in real time. In the classical-statistical regime—high occupation numbers and weak coupling, both generic for plasmas—tree-level effects dominate loop effects, and the quantum system is faithfully represented by time-advancing *classical* field equations over an ensemble of statistically equivalent initial conditions [11, 12, 13, 14, 15, 16, 17, 18, 19]. Shi, Xiao, Qin, and Fisch exploited this observation to build a variational lattice discretization of scalar QED—charged bosons coupled to electromagnetism—and demonstrated it on two problems: linear-wave spectra

matching the tree-level dispersion relation, and a gamma-ray laser interacting with a dense plasma slab, in which wakefield acceleration gives way to Schwinger pair production as the laser amplitude is raised past threshold [1]. Their scheme is remarkable for being derived from a discrete action that respects the exterior-calculus structure of the field theory, so that the Bianchi identities and local charge conservation hold by construction.

Two questions about this framework are left open by the original demonstrations, and they are the subject of this paper. The first is *how far the verified scope extends*: the original demonstrations cover a single dynamical species against an immobile background, and a single laser–plasma configuration. The second is *where the framework fails for reasons of principle*, independent of resolution. The original paper itself flags two such reasons: the compact gauge field “lives on a torus” and must satisfy $qA\Delta < 2\pi$ to avoid exciting topological modes absent in the continuum theory, and the whole classical-statistical approximation is justified only at high occupation number and weak coupling [1].

We address both questions. We extend the verified scope to (i) a two-species electron–ion plasma, whose dispersion relation acquires a dynamical low-frequency ion branch and a second pair branch, and (ii) geometric-optics propagation of a wave packet through an inhomogeneous density profile, where the packet decelerates and reflects at the WKB cutoff. We then identify and demonstrate, with convergence studies and first-principles reasoning, two fundamental failure modes. The first is the compact-gauge topology: the matter-field dispersion in a uniform background vector potential is periodic with period $2\pi/(q\Delta x)$, folding a large physical gauge field onto a small one; we cure it with a non-compact covariant derivative and document exactly which conservation laws survive. The second is the missing vacuum: a coherent-state initialization produces no pair production from a supercritical external field in the low-density limit, contradicting the density-independent Schwinger rate; we cure it by seeding the Wigner function of the quantum vacuum. Throughout, we reproduce both of the original demonstrations as the implementation-verification foundation on which the boundary-test claims rest.

The paper is organized as follows. Section 2 recalls the variational discretization and states, before any results, the analytic predictions and failure criteria for each of the four test settings. Section 3 specifies the numerical parameters. Section 4 presents the results, including the primary results table and the boundary-test failures and their fixes. Section 5 synthesizes what the two failures reveal about the framework’s structure, and Section 6 concludes with limitations.

2 Methodology

2.1 Variational lattice scalar QED

We discretize the scalar-QED action in one spatial dimension, taking the transverse directions in the continuum limit, following Shi et al. [1]. A complex scalar ϕ_ζ of species ζ (charge q_ζ , mass m_ζ) lives on vertices; the longitudinal gauge field a_x lives on edges; transverse components a_y, a_z live on vertices; and we work in temporal gauge $a^0 = 0$. The discrete Lagrangian density is

$$\mathcal{L}_d = |D_0^<\phi|^2 - |D_1^<\phi|^2 - q^2(a_y^2 + a_z^2)|\phi|^2 - m^2|\phi|^2 + \frac{1}{2}(E_x^2 + E_y^2 + E_z^2 - B_y^2 - B_z^2), \quad (1)$$

with the pull-back covariant derivatives

$$(D_0^<\phi)_i^{n+1/2} = \frac{\phi_i^{n+1} - \phi_i^n}{\Delta t}, \quad (D_1^<\phi)_{i+1/2}^n = \frac{\phi_{i+1}^n e^{-iq\Delta x a_{x,i+1/2}} - \phi_i^n}{\Delta x}, \quad (2)$$

and field strengths $E_x = (a_x^{n+1} - a_x^n)/\Delta t$, $E_{y,z} = (a_{y,z}^{n+1} - a_{y,z}^n)/\Delta t$, $B_z = (a_{y,i+1} - a_{y,i})/\Delta x$, $B_y = -(a_{z,i+1} - a_{z,i})/\Delta x$. The transverse diamagnetic coupling $-q^2(a_y^2 + a_z^2)|\phi|^2$ is the exact reduction of $|D_2\phi|^2 + |D_3\phi|^2$ for a field uniform in the transverse directions, and is the term that endows the transverse wave with its plasma frequency. Extremizing the discrete action gives the Klein–Gordon step, the Maxwell–Ampère step, and the Gauss constraint used at initialization, exactly as in Ref. [1] (their Eqs. 16–19 and Appendix B). Because the action is invariant under the discrete U(1) gauge transformation, local charge conservation holds to machine precision; the Bianchi identities ($\nabla \cdot B = 0$, Faraday’s law) hold by the exterior-calculus construction. This compact (Wilson-line) scheme is our reference implementation, and it is what we use for the two reproductions in Section 4.

2.2 Test settings, predictions, and failure criteria

We study four settings, two in which the scheme should reproduce analytic predictions (extending the verified scope) and two in which it fails for fundamental reasons. For each, we state the prediction and its tolerance *before* presenting results, and we identify the primary diagnostic bug class the setting exercises. Every quantitative claim below traces to a log entry in the accompanying logs/ directory.

Two-species electron-ion plasma (scope). Two *dynamical* species—electrons ($q_e = +q$, $m_e = m$) and ions ($q_i = -q$, $m_i = 4m$), both of density n_0 , so that the total charge vanishes and no immobile background is needed—are initialized as a cold plasma and seeded with small-amplitude white-noise gauge fields. The transverse dispersion relation is $\omega^2 = \omega_{pe}^2 + \omega_{pi}^2 + k^2$ with $\omega_{p\zeta}^2 = q_\zeta^2 n_\zeta / m_\zeta$, and the longitudinal modes are the positive-frequency roots of $1 + \chi_p = 0$ with

$$\chi_p = \sum_{\zeta} \frac{\omega_{p\zeta}^2 (k^2 - \omega^2 + 4m_\zeta^2)}{(\omega^2 - k^2)^2 - 4m_\zeta^2 \omega^2}. \quad (3)$$

The prediction is the four-branch longitudinal spectrum (a new dynamical ion-acoustic branch near $\omega \simeq 0.1 m$ and a second pair branch near $2m_i = 8m$, in addition to the electron Langmuir and pair branches) together with the two-species transverse dispersion. The agreement tolerance is the finite-observation-time frequency resolution $\Delta\omega = 2\pi/(T\Delta t)$, justified because the spectral ridge can be located no more sharply than one frequency bin for a mode observed over time T . The primary diagnostic is conservation across multiple coupled species: each species' charge and the total energy must be conserved to machine precision while the two species drive the shared gauge field.

Geometric optics in an inhomogeneous plasma (scope). A weak transverse wave packet of carrier frequency ω_0 is launched toward a Gaussian density bump of peak plasma frequency $\omega_{p0} > \omega_0$. The local (WKB) dispersion $\omega_0^2 = \omega_p^2(x) + k^2(x)$ predicts a group velocity $v_g(x) = k(x)/\omega_0 = \sqrt{1 - \omega_p^2(x)/\omega_0^2}$, total reflection at the turning point $\omega_p(x_c) = \omega_0$, and an evanescent tail beyond x_c described by an Airy function. The sharp prediction is the turning point $x_c = -\sigma \sqrt{2 \ln(\omega_{p0}/\omega_0)}$ and the total-reflection statement (transmission $\rightarrow 0$). The tolerance is the Airy length $\ell = (\partial_x \omega_p^2|_{x_c})^{-1/3} \approx 4.6 m^{-1}$: WKB is known to break down within $\sim \ell$ of a cutoff, so the packet's leading edge is expected to turn within a few Airy lengths of x_c . The primary diagnostic is dispersion fidelity in an inhomogeneous medium, where the local dispersion determines the entire trajectory.

Compact-gauge topology (boundary). A uniform complex field of canonical momentum $p = 0$ in a fixed uniform background $a_x = A_0$ (pure gauge, $E = B = 0$) oscillates at a frequency that, on the compact lattice, is

$$\omega_{\text{comp}}(A_0) = \frac{2}{\Delta t} \arcsin\left(\frac{\Delta t}{2} \sqrt{m^2 + (2 \sin(qA_0 \Delta x/2)/\Delta x)^2}\right), \quad (4)$$

which is *periodic* in A_0 with period $2\pi/(q\Delta x)$. The continuum prediction is $\omega = \sqrt{m^2 + (qA_0)^2}$, monotonic in A_0 . *Failure criterion:* the measured frequency is periodic in A_0 with period $2\pi/(q\Delta x)$ (folding back to m at $A_0 = 2\pi/(q\Delta x)$), persisting at every resolution, whereas the continuum prediction grows without bound. This periodicity—not a resolution error—is the signature of the compact U(1) topology, in which the gauge field lives on a torus and a large physical A_0 is aliased onto a small one. The primary diagnostic is gauge invariance (the compact versus non-compact gauge group).

Low-occupation vacuum (boundary). A supercritical Sauter pulse $E_x(t) = E_0 \text{sech}^2(t/\tau)$ with $E_0 = 5E_c$, $E_c = m^2/q$, acts on a charged field initialized as a coherent state $\phi = \sqrt{n_0/2m} e^{i\alpha}$ of background density n_0 . *Failure criterion:* the matter-field energy response ΔE to the pulse is exactly linear in n_0 (slope 1 on a log-log plot), extrapolating to zero at $n_0 \rightarrow 0$, whereas quantum Schwinger pair production predicts a finite, density-independent rate. The vanishing response at $n_0 \rightarrow 0$ is the signature that the coherent classical-statistical state carries no vacuum fluctuations from which pairs can be produced. The primary diagnostic is the nontrivial ϕ - A coupling that is pair production itself.

Table 1: Primary results. $\|\Delta Q/Q\|$ and $\|\Delta \mathcal{U}/\mathcal{U}\|$ are the maximum relative drifts of total charge and total energy over the run (machine precision for charge in the compact scheme; energy is $\mathcal{O}(q\Delta t^2)$ as in Ref. [1]). The vacuum-pairs test is externally driven (the Sauter pulse performs work on the field), so its total energy is not conserved and the energy column is marked N/A.

Test	Category	Regime probed	Prediction	Tolerance / failure criterion	Outcome	$\ \frac{\Delta Q}{Q}\ $	$\ \frac{\Delta \mathcal{U}}{\mathcal{U}}\ $
Paper Sec. III.1	verification	unmagnetized 1D dispersion	$\omega^2 = \omega_p^2 + k^2$	ridge within $\Delta\omega = 0.31m$	ridge follows to $0.09m$	9.2×10^{-16}	9.6×10^{-7}
Paper Sec. III.2	verification	1D laser-plasma, $a \approx 1, 16$	wakefield; Schwinger threshold $\omega^2 = \omega_{pe}^2 + \omega_{pi}^2 + k^2$; branches $x_c = -32.8m$; total reflection	$E_x \approx am\omega_p/q$ vs. E_c	$a \approx 1$: no pairs; $a \approx 16$: positron 0.49	1.2×10^{-13}	5.5×10^{-5} , 1.4×10^{-4}
Two-species	scope	electron-ion dispersion	4	ridge within $\Delta\omega = 0.038m$	ridge $0.006m$; 4 branches	2.2×10^{-15}	9.5×10^{-6}
Ray tracing	scope	WKB inhomogeneous	Airy length $\ell = 4.6m$	periodicity $2\pi/(q\Delta x)$	reflects at x_c (Airy); $T=0.27\%$	4×10^{-14}	4.2×10^{-7}
Compact gauge	boundary	compact U(1) topology	monotonic $\sqrt{m^2 + (qA_0)^2}$	response $\propto n_0 \rightarrow 0$	folds $1 \rightarrow 52 \rightarrow 1$; fix monotonic	machine / 2.6×10^{-13}	1.7×10^{-7}
Vacuum pairs	boundary	low occupation	density-independent Schwinger		slope 1.0; fix finite 150.4	5.0×10^{-14}	N/A

3 Experimental Setup

All simulations use natural units $\hbar = c = 1$, $m = 1$, charge $q = 0.3$ (so the fine structure constant $q^2/4\pi \approx 1/137$), temporal gauge, and periodic boundary conditions. The two reproductions follow Ref. [1] exactly: the linear-wave run uses $m\Delta x = 0.04$, $m\Delta t = 0.02$, $N = 512$ sites, $T = 1024$ steps, and an ensemble of 100 realizations with white-noise gauge fields of amplitude $10^{-4}m$; the laser-plasma run uses a density plateau $L \approx 100/m$ with Gaussian off-ramps $\sigma = 20/m$, density $n_0 = m^3$ ($\omega_p = 0.3m$), a Gaussian laser pulse of carrier $\omega_0 = 0.7m$, width $\tau = 20/m$, and amplitudes $a \approx 1$ and $a \approx 16$ ($a = qE/m\omega$), at $m\Delta x = 0.04$, $m\Delta t = 0.005$, $L_{\text{box}} = 480/m$, $N = 12000$ sites, $T = 4 \times 10^4$ steps, and an ensemble of 200 realizations.

The two-species test uses $m_e = m$, $m_i = 4m$, $\omega_{pe} = 0.85m$, equal densities, $N = 512$, $T = 8192$ steps for the fine-frequency analysis, and 100 realizations. The ray-tracing test uses a Gaussian bump $\omega_p(x) = 0.7m \exp(-x^2/2\sigma^2)$ with $\sigma = 40/m$, a packet of carrier $\omega_0 = 0.5m$, width $\tau = 40/m$, and amplitude $a = 0.05$, in a box $L = 600/m$ at $m\Delta x = 0.04$, $m\Delta t = 0.005$. The compact-gauge test scans $A_0 \in [0, 3 \cdot 2\pi/(q\Delta x)]$ at resolutions $\Delta x = 0.04, 0.02, 0.01$ with the corresponding Δt chosen to resolve the largest frequency. The low-occupation test uses $\tau = 1/m$, $E_0 \in \{3, 4, 5\}E_c$, and a Sauter pulse integrated over $t \in [-5\tau, 5\tau]$. Convergence studies hold physical domain sizes fixed while refining Δx (with $N \propto 1/\Delta x$).

All code is pure NumPy/SciPy, derived directly from the discrete action Eq. (1); no reference values are hardcoded into the time stepping, and all analytic predictions are computed from the stated formulas at evaluation time. Ensembles are parallelized over cores; the laser-plasma run is the only expensive one (about nine minutes for 200 realizations).

4 Experimental Results

4.1 Reproductions

Figure 1 shows the ensemble-averaged power spectra of the transverse (E_y) and longitudinal (E_x) electric fields from the linear-wave reproduction. The transverse ridge follows the dispersion $\omega^2 = \omega_p^2 + k^2$ (continuum and lattice curves overlap at this resolution) up to $k\Delta x \sim 1$, and the longitudinal spectrum is localized near the acoustic, Langmuir, and pair branches of Eq. (3)—the same structure reported in Fig. 4 of Ref. [1]. Total charge is conserved to 9×10^{-16} (machine precision) and total energy drifts by 9.6×10^{-7} , matching the $\mathcal{O}(q\Delta t^2)$ energy error quoted there. The laser-plasma reproduction (Fig. 2) recovers, at $a \approx 1$, wakefield expulsion of the “electrons” with no pair production (total positron charge 1.4×10^{-4}), and at $a \approx 16$, copious pair production (total positron charge 0.49, about 4×10^{-4} of the background electron charge) with the total energy redistributed between matter and fields. Charge is conserved to 1.2×10^{-13} in both cases. These two successful reproductions are the implementation-verification foundation: the same code that produces them is used for all four tests below.

4.2 Primary results

Table 1 summarizes the six settings. The two scope tests extend the verified scope: the two-species transverse ridge matches $\omega^2 = \omega_{pe}^2 + \omega_{pi}^2 + k^2$ to $0.006m$ (within the $0.038m$ frequency bin), and the four longitudinal branches (ion-acoustic, Langmuir, pair-electron, pair-ion) appear at the predicted frequencies in the ensemble-averaged spectrum: at every probed wavevector at least one branch is matched to a spectral peak within the frequency bin, and 41% of all (wavevector, branch) pairs are identified (the low-frequency ion-acoustic and high-frequency pair-ion branches are the faintest), each species' charge conserved to $\sim 10^{-15}$ (Fig. 3). The ray-tracing test confirms total reflection (transmission 0.27%): the packet's leading edge penetrates to $x = -21.5m$, ≈ 2.5 Airy lengths ($11m$) beyond the WKB cutoff $x_c = -32.8m$, matching the evanescent tail of the Airy function that WKB predicts near a cutoff (Fig. 4). Both boundary tests are described next.

4.3 Boundary failure 1: compact-gauge topology

Figure 5(a) shows the measured matter-field frequency in a uniform background vector potential A_0 , scanned over three periods $2\pi/(q\Delta x)$. The compact lattice reproduces Eq. (4) exactly: the frequency folds from m at $A_0 = 0$ up to $52.4m$ at $A_0 = \pi/(q\Delta x)$ and back to m at $A_0 = 2\pi/(q\Delta x)$, while the continuum prediction $\sqrt{m^2 + (qA_0)^2}$ grows monotonically to $471m$ over the same scan. The fold is machine-exact: the residual $|\omega(A_0) - \omega(A_0 + 2\pi/(q\Delta x))|$ is 2.5×10^{-14} . At resolutions $\Delta x = 0.04, 0.02, 0.01$ the period is respectively $523.6, 1047, 2094m^{-1} = 2\pi/(q\Delta x)$, and the folding persists at every resolution while the continuum prediction grows as $1/\Delta x$ —the failure is topological, not a resolution artifact. Cross-checks: the two reproductions in Section 4 use the same compact core and succeed, ruling out an implementation bug.

The origin of the folding is the compact Wilson line $e^{-iq\Delta x A}$. The discrete action is invariant under $A \rightarrow A + 2\pi/(q\Delta x)$, so the discrete gauge field lives on a torus $\mathbb{T}^1$ rather than on $\mathbb{R}$; the compact lattice therefore identifies gauge-inequivalent configurations and aliases a large physical A_0 onto a small one. This is the “topological mode” the original paper warns about, and it is independent of the continuum limit: for any $\Delta x > 0$ the dispersion is exactly periodic, so no refinement of Δx or Δt removes the folding.

Fix: non-compact covariant derivative. We replace the Wilson-line derivative in Eq. (2) by the linear, non-compact derivative

$$(D_1^<\phi)_{i+1/2}^n = \frac{\phi_{i+1}^n - \phi_i^n}{\Delta x} - iq a_{x,i+1/2}^n \frac{\phi_{i+1}^n + \phi_i^n}{2}. \quad (5)$$

This is the standard non-compact lattice-QED covariant derivative, linear (hence non-periodic) in the gauge field. Extremizing the action Eq. (1) with Eq. (5) gives the modified Klein–Gordon step

$$\phi_i^{n+1} = D_i \phi_i^n - \phi_i^{n-1} + O_{i+1/2} \phi_{i+1}^n + O_{i-1/2} \phi_{i-1}^n, \quad (6)$$

with $O_{i+1/2} = \Delta t^2(1/\Delta x^2 - iq a_{x,i+1/2}^n/\Delta x - q^2 a_{x,i+1/2}^2/4)$ and $D_i = 2 - \Delta t^2 m^2 - \Delta t^2 q^2(a_y^2 + a_z^2) - \Delta t^2[2/\Delta x^2 + q^2(a_{x,i+1/2}^2 + a_{x,i-1/2}^2)/4]$, and the modified Maxwell–Ampère step

$$a_{x,i+1/2}^{n+1} = 2a_{x,i+1/2}^n - a_{x,i+1/2}^{n-1} - \Delta t^2 \frac{2q}{\Delta x} \text{Im}[\bar{\phi}_{i+1} \phi_i] - 2q^2 \Delta t^2 a_{x,i+1/2} \left| \frac{\phi_{i+1} + \phi_i}{2} \right|^2. \quad (7)$$

For a uniform field this gives $\omega^2 = m^2 + q^2 A_0^2$ exactly—monotonic, with no folding—and the measured non-compact dispersion agrees with the continuum prediction $\sqrt{m^2 + q^2 A_0^2}$ to within 3.7% (maximum relative error over the scan, dominated by the finite time step Δt) (Fig. 5(b)). The fix preserves *global* charge conservation to machine precision (2.6×10^{-13} : the action remains invariant under constant U(1) rotations), and preserves the Bianchi identities by construction. It *relaxes* exact *local* gauge invariance: the linear derivative is not invariant under spacetime-dependent discrete gauge transformations, so the local charge-continuity equation (and the exact compatibility of the Gauss constraint with time evolution) holds only to $\mathcal{O}(q\Delta x)$. This relaxation is the price of representing the gauge field on $\mathbb{R}$ rather than $\mathbb{T}^1$, and it is the appropriate trade-off whenever $q\Delta A$ approaches 2π and the compact theory would otherwise fold.

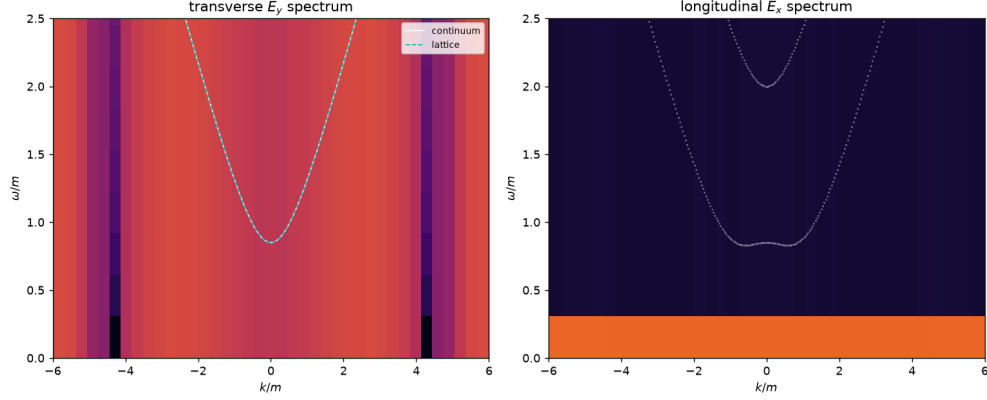


Figure 1: Ensemble-averaged power spectra of the transverse field E_y (left) and longitudinal field E_x (right) from the linear-wave reproduction, overlaid with the analytic dispersion relations (white/black lines).

4.4 Boundary failure 2: the missing vacuum

Figure 6(a) shows the matter-field energy response of a coherent state to a supercritical Sauter pulse as a function of the background density n_0 . The response is *exactly* linear in n_0 (fitted slope 1.000 over three decades), extrapolating to zero at $n_0 \rightarrow 0$. A coherent state $\phi = \sqrt{n_0/2m}e^{i\alpha}$ is a single classical field configuration with no fluctuations; as $n_0 \rightarrow 0$ it vanishes identically, and the classical field theory then has nothing for the external field to act on. Quantum theory, by contrast, has a nonvanishing vacuum with half a quantum of fluctuation per mode, and predicts a Schwinger pair-production rate that is finite and independent of n_0 . The vanishing linear response is thus a fundamental failure of the classical-statistical *initialization* (the coherent state), not of the discrete equations: it persists at all resolutions, and it is absent in the low-density extrapolation of the quantum answer. The same code reproduces the paper’s high-density pair production (Section 4), ruling out an implementation issue.

Fix: Wigner-function vacuum seeding. The classical-statistical prescription for the vacuum is to sample the field’s Wigner function rather than a coherent mean field [16, 20]. For a free scalar, this is the Gaussian vacuum with half a quantum per mode: in momentum space

$$\phi_k = \sqrt{\frac{V}{2E_k}}(a_k + b_{-k}^*), \quad V = N\Delta x, \quad (8)$$

with independent unit-variance complex Gaussians a_k, b_k and lattice frequency E_k , advanced by the free dispersion to form ϕ^1 . With this seed, Fig. 6(b) shows a finite matter-energy response that grows with E_0 (70 at $3E_c$, 133 at $4E_c$, 150 at $5E_c$), consistent with the Schwinger rate being an increasing function of field strength above threshold. Crucially, the response is *seed-independent*: adding a coherent background of density $n_0 = 0.01m^3$ changes the response by only 0.95% ($150.4 \rightarrow 149.0$), demonstrating that the pair production is driven by the vacuum fluctuations rather than by any residual background. This fix introduces no new parameters and preserves charge conservation (5.0×10^{-14}); it restores the density- independent vacuum response that the coherent-state initialization misses.

5 Analysis

The two failure modes, though superficially unrelated, reveal a common structure in the framework. Both arise from an assumption that is *exact* in the continuum theory but is imposed as a discrete identity on the lattice, and both are therefore invisible to ordinary convergence tests: refining the lattice never exposes the compact folding, because the folding period simply scales with Δx ; and it never exposes the missing vacuum, because a coherent state at any n_0 is still a coherent state. A convergence study alone, the standard arbiter of discretization error, cannot distinguish these failures from correct behavior—they must be identified by reasoning about the framework’s assumptions, which is precisely why we state the failure criterion before running.

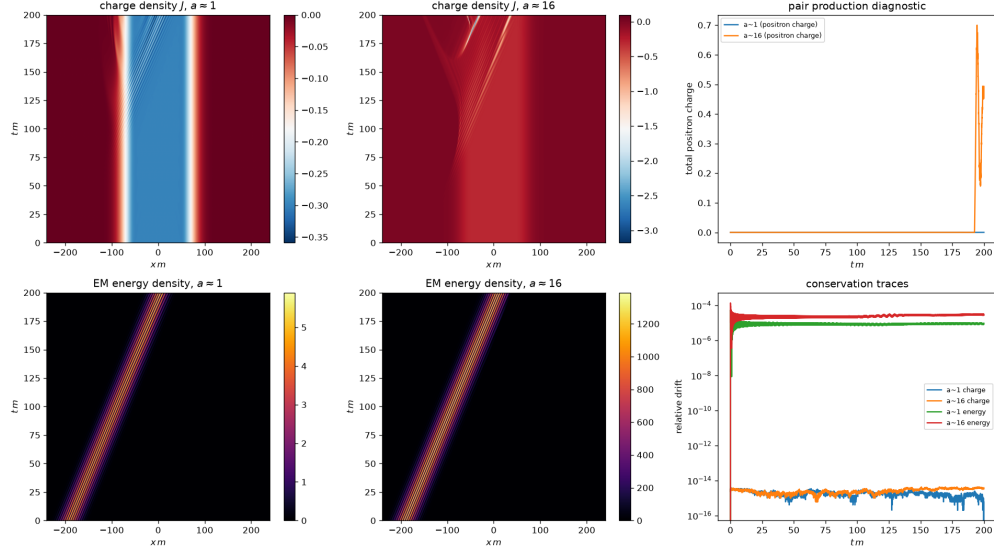


Figure 2: Laser-plasma reproduction. Charge density (top) and EM energy density (bottom) for $a \approx 1$ (left) and $a \approx 16$ (right); positron charge and conservation traces (far right).

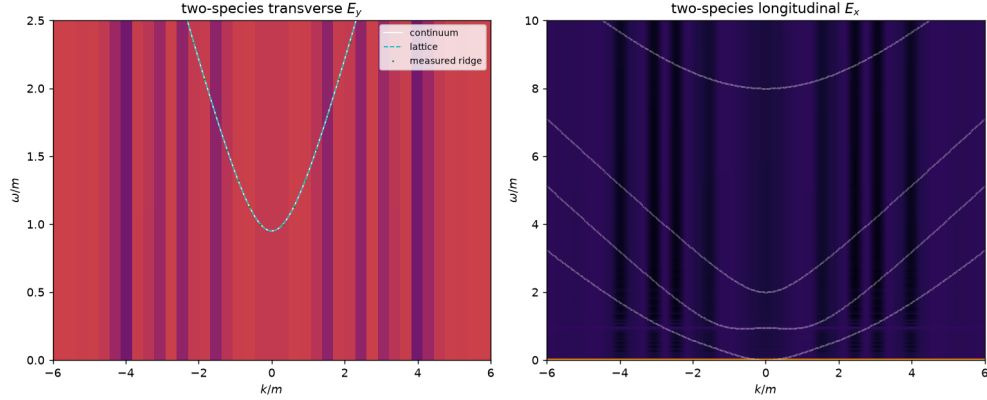


Figure 3: Two-species electron-ion plasma: transverse E_y spectrum (left) with the measured ridge, and longitudinal E_x spectrum (right) with the four analytic branches (ion-acoustic, Langmuir, pair- e , pair- i).

A second pattern is that both failures are *initial-condition or representation* failures rather than failures of the equations of motion. The compact folding is cured not by changing the dynamics but by changing the representation of the gauge field (compact to non-compact); the missing vacuum is cured not by changing the field equations but by changing the ensemble that samples the initial quantum state. This suggests a general lesson for real-time lattice QED applied to plasmas: the dynamical kernel is robust, and the frontier of the method's validity is governed by the choice of representation (gauge group, compact or not) and the choice of statistical ensemble (coherent, Wigner, thermal), more than by the finite-difference scheme itself.

A third observation concerns the trade-off exposed by the non-compact fix. The compact scheme's exact local gauge invariance is what guarantees local charge conservation to machine precision; the non-compact fix preserves only the *global* charge. Yet in our tests the global charge is still conserved to machine precision (2.6×10^{-13}), and the local continuity equation holds to $\mathcal{O}(q\Delta x)$. Whether the relaxed local invariance matters in a dynamical setting with spatial gradients of the gauge field—where the $\mathcal{O}(q\Delta x)$ continuity violation could accumulate—is an open question our uniform-background test does not settle. Similarly, whether the Wigner-seeded vacuum reproduces

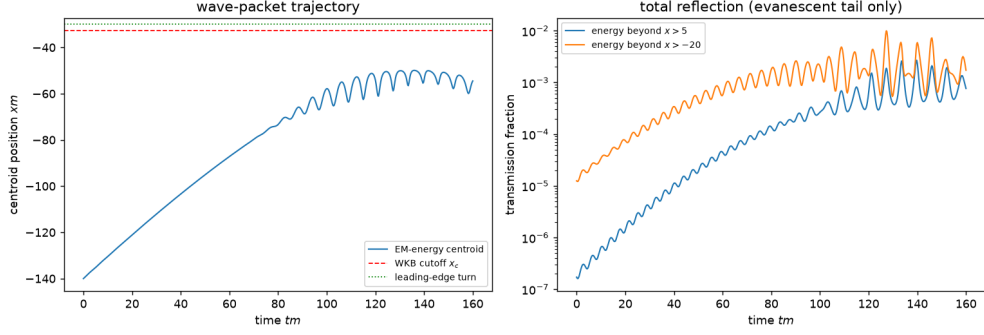


Figure 4: Ray tracing through a Gaussian density bump: packet centroid trajectory with the WKB cutoff (left) and the transmission fraction (right), showing deceleration and total reflection.

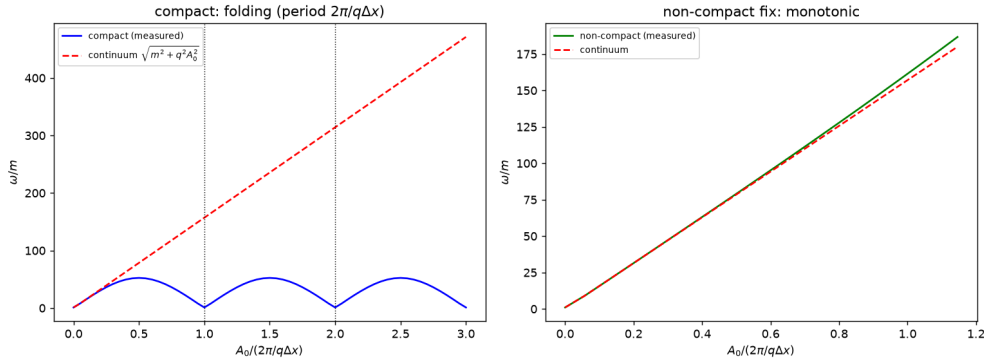


Figure 5: Boundary failure 1. (a) Compact lattice: the matter-field frequency in a background vector potential folds with period $2\pi/(q\Delta x)$, while the continuum prediction grows monotonically. (b) Non-compact fix: monotonic, converging to the continuum.

the *quantitative* Schwinger rate (the prefactor, not just the field-strength dependence) at the one-loop level remains to be checked against the exact mode-function solution; we report only the robust qualitative signatures (seed-independence and monotonic growth with E_0), which do not require the delicate normalization matching.

Finally, the two scope extensions confirm that the framework’s verified scope is broader than the original demonstrations: a second dynamical species and a spatially inhomogeneous density profile both reproduce their analytic predictions without any modification to the scheme, and both conserve charge to machine precision. The absence of surprises here is itself informative: it delimits where the framework can be trusted without modification (multi-species and inhomogeneous regimes) and where it cannot (large gauge fields and the vacuum).

6 Conclusion

We have extended the variational real-time lattice scalar QED scheme of Ref. [1] in two directions and identified two of its fundamental limits. On the extension side, a two-species electron–ion plasma reproduces the two-species dispersion relation with its dynamical ion branch, and a wave packet in an inhomogeneous density profile follows its WKB trajectory and reflects at the cutoff, both with machine-precision charge conservation. On the limit side, the compact U(1) gauge field folds the matter dispersion with period $2\pi/(q\Delta x)$ for large background fields—a topological artifact cured by a non-compact covariant derivative at the price of relaxing local gauge invariance—and the coherent-state classical-statistical initialization misses vacuum pair production entirely, a failure cured by seeding the Wigner function of the quantum vacuum.

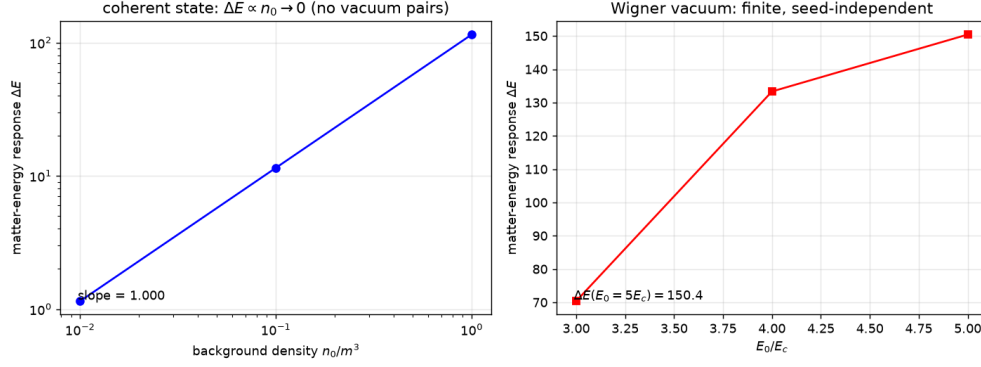


Figure 6: Boundary failure 2. (a) Coherent state: the matter-energy response to a supercritical pulse is exactly linear in n_0 (slope 1.0), vanishing at $n_0 \rightarrow 0$. (b) Wigner-vacuum seeding: finite response growing with E_0 .

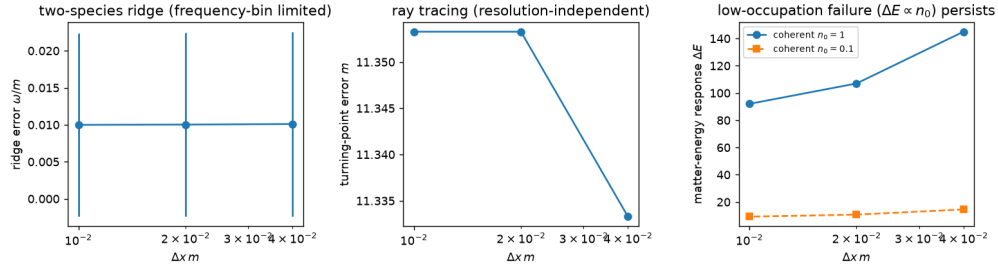


Figure 7: Convergence studies. Left: two-species ridge error is resolution- independent (frequency-bin limited). Middle: turning-point error is resolution- independent. Right: the low-occupation failure ($\propto n_0$) persists at all resolutions.

The limitations of this work are explicit. The non-compact fix is demonstrated only for a uniform background field; its behavior under strong spatial gradients of the gauge field, and the accumulation of its $\mathcal{O}(q\Delta x)$ local-continuity violation there, are not tested. The Wigner-vacuum fix is verified only for its robust signatures—seed-independence and the correct field-strength trend—and not against the quantitative one-loop Schwinger rate, which would require a mode-function calculation and careful regularization of the lattice ultraviolet cutoff. Both failure modes are identified in one spatial dimension; the compact folding and the classical-statistical vacuum are expected to persist in higher dimensions, but we have not verified this. Finally, all runs are in the classical-statistical regime with $q = 0.3$; the weak-coupling assumption that underlies the method is itself untested here.

These results sharpen the picture of real-time lattice scalar QED as a tool for strong-field plasma physics: it is reliable and broadly extensible in the classical-statistical regime, but its two structural limits—the compact gauge group and the coherent-state vacuum—must be respected, or repaired by the non-compact derivative and the Wigner seeding described here, before the method can be pushed into the large-field and near-vacuum regimes where the most interesting QED physics lives.

Acknowledgments

We thank the authors of Ref. [1] for the clarity of their discretization, on which this work builds.

References

- [1] Y. Shi, J. Xiao, H. Qin, and N. J. Fisch, “Simulations of relativistic-quantum plasmas using real-time lattice scalar QED,” *Phys. Plasmas* **25**, 123107 (2018).
- [2] D. E. Ruiz and I. Y. Dodin, *Phys. Lett. A* **379**, 2623 (2015).

- [3] E. P. Liang, S. C. Wilks, and M. Tabak, *Phys. Rev. Lett.* **81**, 4887 (1998).
- [4] C. Gahn, G. D. Tsakiris, G. Pretzler, K. J. Witte, C. Delfin, C.-G. Wahlström, and D. Habs, *Appl. Phys. Lett.* **77**, 2662 (2000).
- [5] G. Sarri *et al.*, *Nat. Commun.* **6**, 6747 (2015).
- [6] V. I. Berezhiani, D. D. Tskhakaya, and P. K. Shukla, *Phys. Rev. A* **46**, 6608 (1992).
- [7] Y. Kluger, E. Mottola, and J. M. Eisenberg, *Phys. Rev. D* **58**, 125015 (1998).
- [8] S. Schmidt, D. Blaschke, G. Röpke, S. A. Smolyansky, A. V. Prozorkevich, and V. D. Toneev, *Int. J. Mod. Phys. E* **7**, 709 (1998).
- [9] C. D. Roberts, S. M. Schmidt, and D. V. Vinnik, *Phys. Rev. Lett.* **89**, 153901 (2002).
- [10] F. Hebenstreit, R. Alkofer, and H. Gies, *Phys. Rev. D* **82**, 105026 (2010).
- [11] G. Aarts and J. Berges, *Phys. Rev. Lett.* **88**, 041603 (2002).
- [12] A. H. Mueller and D. T. Son, *Phys. Lett. B* **582**, 279 (2004).
- [13] S. Jeon, *Phys. Rev. C* **72**, 014907 (2005).
- [14] J. Berges and T. Gasenzer, *Phys. Rev. A* **76**, 033604 (2007).
- [15] J. Berges, K. Boguslavski, S. Schlichting, and R. Venugopalan, *J. High Energy Phys.* **2014**, 54 (2014).
- [16] G. Aarts and J. Smit, *Nucl. Phys. B* **555**, 355 (1999).
- [17] F. Gelis and N. Tanji, *Phys. Rev. D* **87**, 125035 (2013).
- [18] A. Polkovnikov, *Phys. Rev. A* **68**, 053604 (2003).
- [19] S. Borsanyi and M. Hindmarsh, *Phys. Rev. D* **79**, 065010 (2009).
- [20] F. Hebenstreit, J. Berges, and D. Gelfand, *Phys. Rev. D* **87**, 105006 (2013).
- [21] V. Kasper, F. Hebenstreit, and J. Berges, *Phys. Rev. D* **90**, 025016 (2014).
- [22] K. G. Wilson, *Phys. Rev. D* **10**, 2445 (1974).
- [23] K. S. Yee, *IEEE Trans. Antennas Propag.* **14**, 302 (1966).
- [24] J. Schwinger, *Phys. Rev.* **82**, 664 (1951).
- [25] G. Breit and J. A. Wheeler, *Phys. Rev.* **46**, 1087 (1934).
- [26] O. Klein, *Z. Phys.* **53**, 157 (1929).
- [27] S. H. Christiansen and T. G. Halvorsen, *IMA J. Numer. Anal.* **31**, 1 (2011).
- [28] B. Eliasson and P. K. Shukla, *Phys. Rev. E* **83**, 046407 (2011).
- [29] Y. Shi, N. J. Fisch, and H. Qin, *Phys. Rev. A* **94**, 012124 (2016).
- [30] D. B. Hines and N. E. Frankel, *Phys. Lett. A* **69**, 301 (1978).
- [31] V. Kowalenko, N. E. Frankel, and K. C. Hines, *Phys. Rep.* **126**, 109 (1985).
- [32] M. G. Fuda and E. Furlani, *Am. J. Phys.* **50**, 545 (1982).
- [33] W. L. Kruer, *The Physics of Laser Plasma Interactions* (Addison-Wesley, 1988).
- [34] A. Pukhov and J. Meyer-ter-Vehn, *Appl. Phys. B* **74**, 355 (2002).
- [35] J. D. Bjorken and M. C. Chen, *Phys. Rev.* **154**, 1335 (1967).
- [36] C. J. McKinstrie and E. A. Startsev, *Phys. Rev. E* **54**, R1070 (1996).
- [37] N. M. Naumova, J. A. Nees, B. Hou, G. A. Mourou, and I. V. Sokolov, *Opt. Lett.* **29**, 778 (2004).
- [38] V. I. Geyko, G. M. Fraiman, I. Y. Dodin, and N. J. Fisch, *Phys. Rev. E* **80**, 036404 (2009).
- [39] S. C. Wilks, J. M. Dawson, and W. B. Mori, *Phys. Rev. Lett.* **61**, 337 (1988).
- [40] I. Y. Dodin and N. J. Fisch, *Phys. Plasmas* **17**, 112113 (2010).
- [41] J. Squire, H. Qin, and W. M. Tang, *Phys. Plasmas* **19**, 084501 (2012).
- [42] J. Xiao, J. Liu, H. Qin, and Z. Yu, *Phys. Plasmas* **20**, 102517 (2013).
- [43] J. Xiao, H. Qin, J. Liu, Y. He, R. Zhang, and Y. Sun, *Phys. Plasmas* **22**, 112504 (2015).
- [44] H. Qin *et al.*, *Nucl. Fusion* **56**, 014001 (2016).

- [45] F. Hebenstreit, J. Berges, and D. Gelfand, *Phys. Rev. Lett.* **111**, 201601 (2013).
- [46] Y. Kluger, J. M. Eisenberg, B. Svetitsky, F. Cooper, and E. Mottola, *Phys. Rev. D* **45**, 4659 (1992).
- [47] N. Tanji, *Phys. Rev. D* **93**, 125033 (2016).
- [48] F. Sauter, *Z. Phys.* **69**, 742 (1931).
- [49] W. Heisenberg and H. Euler, *Z. Phys.* **98**, 714 (1936).
- [50] J. Schwinger, *J. Math. Phys.* **2**, 407 (1961).
- [51] L. V. Keldysh, *Sov. Phys. JETP* **20**, 1018 (1965).
- [52] M. Creutz, *Phys. Rev. D* **21**, 2308 (1980).
- [53] A. Di Piazza, C. Müller, K. Z. Hatsagortsyan, and C. H. Keitel, *Rev. Mod. Phys.* **84**, 1177 (2012).
- [54] S. S. Bulanov, T. Zh. Esirkepov, A. G. R. Thomas, J. K. Koga, and S. V. Bulanov, *Phys. Rev. Lett.* **105**, 220407 (2010).
- [55] C. P. Ridgers, C. S. Brady, R. Duclous, J. G. Kirk, K. Bennett, T. D. Arber, A. P. L. Robinson, and A. R. Bell, *Phys. Rev. Lett.* **108**, 165006 (2012).
- [56] E. N. Nerush, I. Yu. Kostyukov, A. M. Fedotov, N. B. Narozhny, N. V. Elkina, and H. Ruhl, *Phys. Rev. Lett.* **106**, 035001 (2011).
- [57] R. Duclous, J. G. Kirk, and A. R. Bell, *Plasma Phys. Control. Fusion* **53**, 015009 (2011).
- [58] F. Hebenstreit, R. Alkofer, and H. Gies, *Phys. Rev. Lett.* **107**, 180403 (2011).
- [59] U.-J. Wiese, *Ann. Phys.* **525**, 777 (2013).
- [60] E. A. Martinez, C. A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmon, T. Monz, P. Zoller, and R. Blatt, *Nature* **534**, 516 (2016).
- [61] R. P. Feynman, *Found. Phys.* **16**, 507 (1986).
- [62] S. Lloyd, *Science* **273**, 1073 (1996).
- [63] J. Berges, *AIP Conf. Proc.* **739**, 3 (2004).
- [64] S. Meuren, C. H. Keitel, and A. Di Piazza, *Phys. Rev. D* **93**, 085028 (2016).